
MODEL SELECTION AND VALIDATION REPORT (remove_9_biomass model)

OBSERVATIONS:1681

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_9_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.13	0.17
remove_9_tric	0.55	0.03
remove_9_teve	-0.42	0.03
sbt_mean	0.37	0.09
sbt_sd	0.04	0.05
sst_sd	0.06	0.04
log_chla_mean	0.06	0.04
depth	0.21	0.05
log_sum_fishing	0.07	0.03

Dispersion parameter: 0.96

Spatiotemporal AR1 correlation (rho): 0.92

Matérn range: 209.34

Spatiotemporal marginal AR1 SD: 0.84

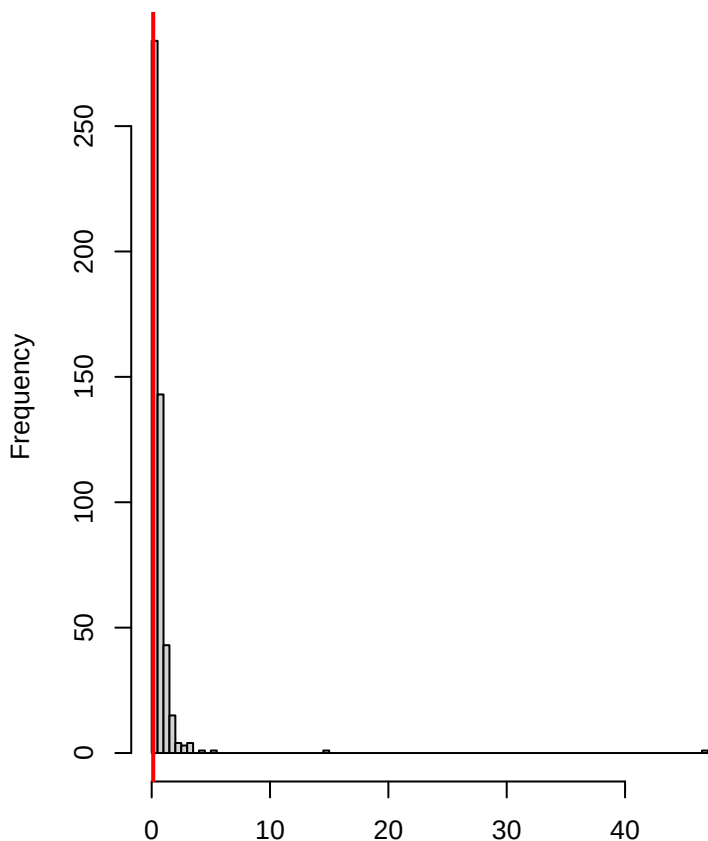
REML criterion at convergence: 10264.876

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

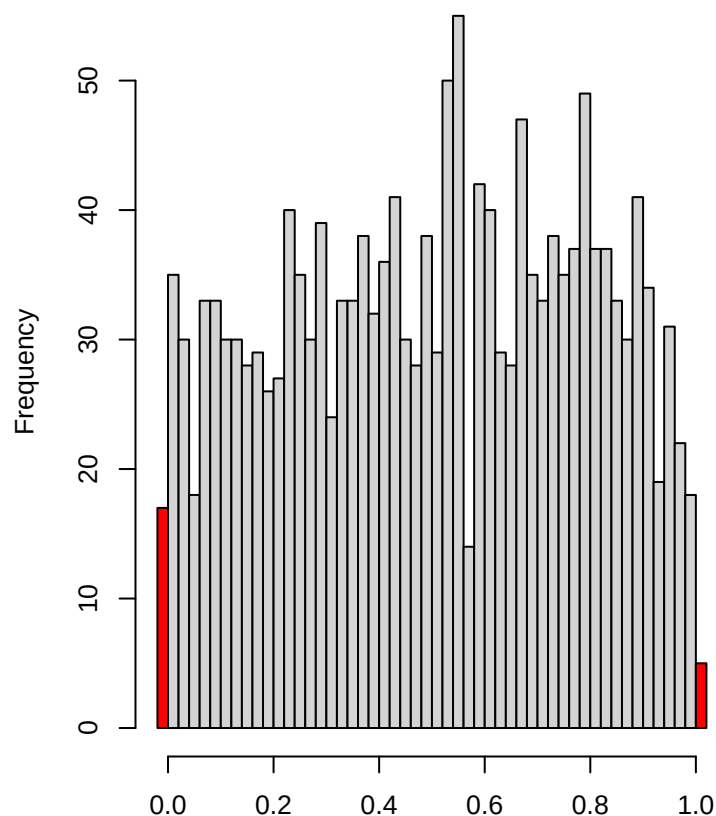
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	20555.75	2.108304
spatiotemporal_random_fields_ar1_with_anisotropy	15	20557.86	75.451405
spatial_random_fields	12	20633.31	2.980941
spatial_random_fields_with_anisotropy	14	20636.29	100.237719
spatiotemporal_random_fields_iid_with_anisotropy	14	20736.53	1.375419
spatiotemporal_random_fields_iid	12	20737.91	153.272869
error_only	10	20891.18	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



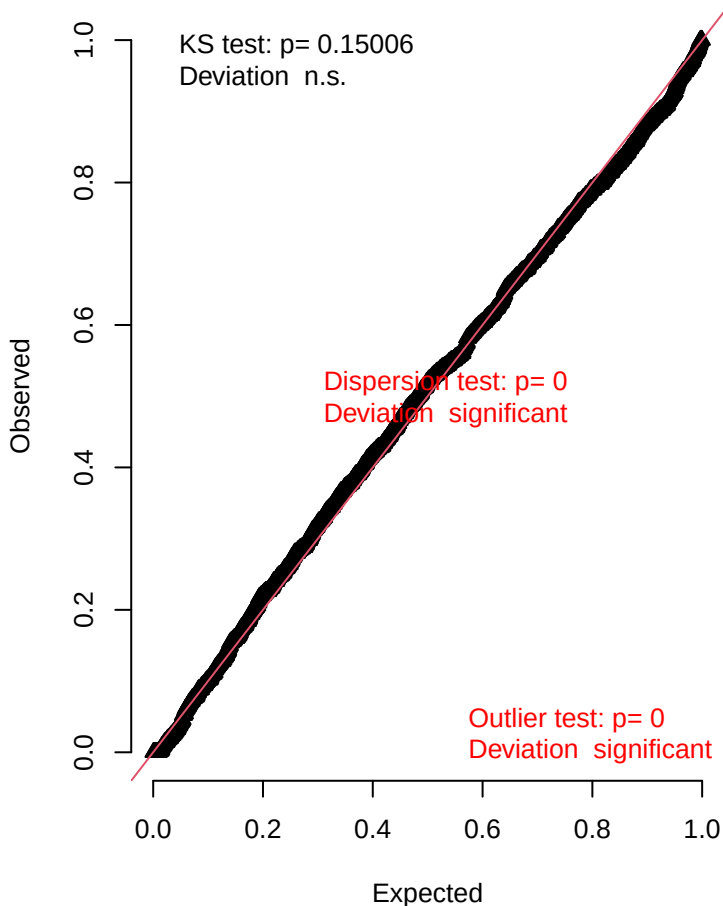
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

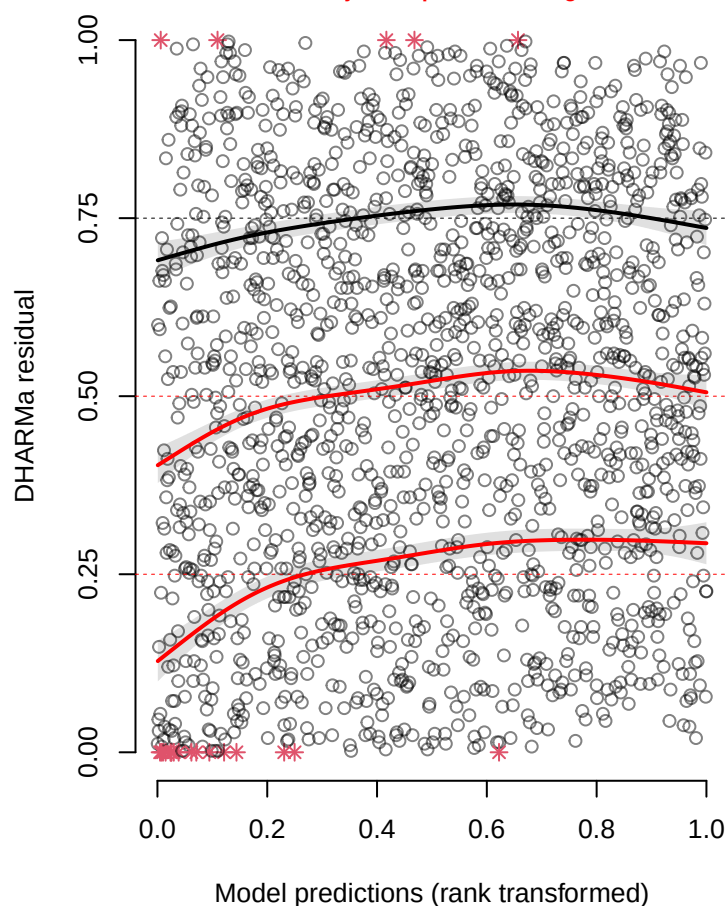


Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation

Moran's I = 0.014 | H1: Distance-based autocorrelation | p-value = 0.112

